RISHIKESH SAROJ

B.Tech Mechanical Engineering · Software & AI/ML Engineer

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PDPM IIITDM Jabalpur, India · Expected graduation 2028 · Open to relocation — seeking internship / full-time roles in Japan

Professional Summary

Mechanical engineering undergraduate building end-to-end ML and software applications (FastAPI, PostgreSQL, React, Docker, Scikit-learn, LLMs/OCR). CampusPilot serves ~1,500 students and is expanding to mobile via React Native. Strong full-stack fundamentals with 60+ automated tests and containerized deployments.

Seeking a software engineering / AI-ML internship or full-time role in Japan. Eager to contribute to a product-driven engineering team and grow within a Japanese company long-term.

Technical Skills

Languages	Python (primary) · JavaScript/TypeScript · SQL · C++
AI / ML	Scikit-learn · (Phi-3, Mistra
Backend	FastAPI · REST APIs · SQLModel · PostgreSQL · Alembic · Docker · pytest
Frontend Tools	React · Vite

Projects

CampusPilot — Academic management platform (~1,500 students)

In progress · github.com/rishisf12/CampusPilot

Python · FastAPI · Hugging Face (LLMs) · Tesseract OCR · ReportLab · React

- Built an OCR + LLM pipeline that converts exam seating PDFs into JSON for roll-number lookup and generates personalized PDF timetables.
- Built attendance tracking with Safe/Warning/Critical flags, a “classes you can miss” calculator, and live current/next class views from PDF/CSV timetables.
- Designed a layered architecture (routes, services, models, OCR modules) backed by 63+ unit tests.

pocketF — Student budgeting app (solo project)

In progress · github.com/rishisf12

Python · FastAPI · PostgreSQL · Pandas · Matplotlib · Docker Compose · pytest · React

- Built a CSV importer using Decimal for exact money math, flagging bad rows across 9 error types.
- Built burn-rate analytics and a “money runs out on” forecast with SAFE/ALERT verdicts and auto-generated charts.
- Containerized API + PostgreSQL with Docker Compose; tested forecast edge cases with pytest.

UrTeam — Hackathon matchmaking platform

github.com/rishisf12

Python · FastAPI · SQLModel · PostgreSQL/SQLite · React · Vite · Tailwind CSS

- Matched students into hackathon teams via skill overlap using set-based Jaccard similarity (no ML dependencies).
- Designed 15+ REST endpoints (student CRUD with 409 duplicate detection, ?tech= idea filter), domain-validated emails, SQLite/PostgreSQL dual-database with auto table creation.
- Built a chat approval flow with owner-verified access control; 8 reusable React components with inline Tailwind CSS.

Master-DSA — DSA practice & language comparison repo

github.com/rishisf12/master-dsa

C++ · Python · Git/GitHub

- Practicing DSA problems and tracking learning progress across topics.
- Comparing C++ and Python code execution, performance, and limits side by side.
- Maintaining a personal DSA learning log with implemented solutions and notes.

Education

PDPM IIITDM Jabalpur, Madhya Pradesh, India

B.Tech, Mechanical Engineering · Expected 2028 · CPI: 6.9 / 10

Class XII: 80% | Class X: 92%

Achievements

- **Smart India Hackathon 2026** — Team of 6; proposed SkyGuard AI (SIH26073), Isolation Forest + SHAP anomaly detection to separate genuine weather events from sensor faults in Automatic Weather Stations.
- **Patent** — Co-inventor, Indian Patent Application No. 202621096766 A: adjustable hybrid welding device with integrated safety measures.
- **Jagrati (Volunteer)** — Teach underprivileged village children weekly; help plan the student-run initiative's activities.

References available upon request. · Portfolio & code: github.com/rishisf12